1.7 Solving Equations with Variables on Both Sides

Name _____ Period ____ Date _____

Algebra I 1.7 (30-minute plan) · Student sheet

I can…

- solve equations with variables on both sides.
- identify equations that have no solution or infinitely many solutions.
- write and solve these equations for real-life problems.

Vocabulary · fill in the missing word

Variables on both sides — move the variable _____ to one side first

Identity — an _____ true for every value of the variable (infinitely many solutions)

No solution — the equation _____ to a false statement, such as 3 = 7

Part 1 · Notes and examples (fill in each note, then work the example)

Variables on both sides: Collect the variable terms on _____ side.

Example 1

Solve 7x + 4 = 3x − 12.

Identity: Variables cancel and a true statement is left: _____.

No solution: Variables cancel and a false statement is left: _____.

Example 2

Solve each equation.

a. 2(x + 3) = 2x + 6 b. 3x + 5 = 3x − 1

Part 2 · Partner practice

Partner practice

Solve 4(x − 2) = 2x + 10.